

GENE EXPRESSION & DISEASE RELATIONSHIP ANALYSIS

1. Introduction

This project analyzes patient-level gene expression data to explore how Gene X and Gene Y expression relate to disease status, smoking status and treatment response. The analysis follows an end-to-end workflow: data preparation in Excel, validation and analysis in MySQL, additional PivotTable analysis in Excel, and an interactive Power BI dashboard to communicate the results.

Objective

To identify observable patterns between gene expression levels and patient characteristics, and to present those patterns clearly to a non-technical audience.

2. Dataset

The dataset is a public collection of 1,000 patient records. Variables include Patient ID, age, gender, Gene X expression, Gene Y expression, smoking status, disease status and treatment response. Several categorical variables were stored as numeric codes, so readable label columns were added while the original coded columns were retained.

Group	Patients
Healthy	962
Disease A	18
Disease B	20
Total	1,000

Overall average expression across all patients was 3.92 for Gene X and 3.63 for Gene Y. The disease groups are small compared with the Healthy group, which should be kept in mind when reading the results.

3. Methodology

Step 1: Data preparation (Excel)

Readable label columns were created for gender, smoking status, disease status and treatment response. Original coded values were kept to preserve data integrity.

Step 2: Data validation (MySQL)

Before analysis, the data was checked for record count, duplicate Patient IDs, missing values in both original and label columns, and column data types.

Step 3: Exploratory and analytical SQL

Queries used aggregate functions (COUNT, AVG, MIN, MAX), GROUP BY, filtering, CASE expressions and subqueries. Analyses included:

- Patient distribution by gender, disease, smoking status and treatment response
- Average gene expression across disease, smoking and treatment-response groups

- Patients with above-average Gene X and Gene Y expression and their disease distribution
- Treatment response across disease and smoking groups

Step 4: Excel PivotTables

PivotTables and PivotCharts summarized the cleaned dataset as a second analytical view and helped confirm that patterns were consistent with the SQL results.

Step 5: Power BI dashboard

An interactive dashboard was built with KPI cards, slicers (disease, smoking status, treatment response, gender), a scatter plot of Gene X against Gene Y, and charts for expression, disease split, smoking status and treatment response.

4. Findings

4.1 Gene expression by disease group

Both genes show higher average expression in the disease groups than in the Healthy group. Disease B has the highest observed average for both genes.

Group	Avg Gene X	Avg Gene Y
Healthy	3.79	3.55
Disease A	6.40	4.58
Disease B	7.94	6.52

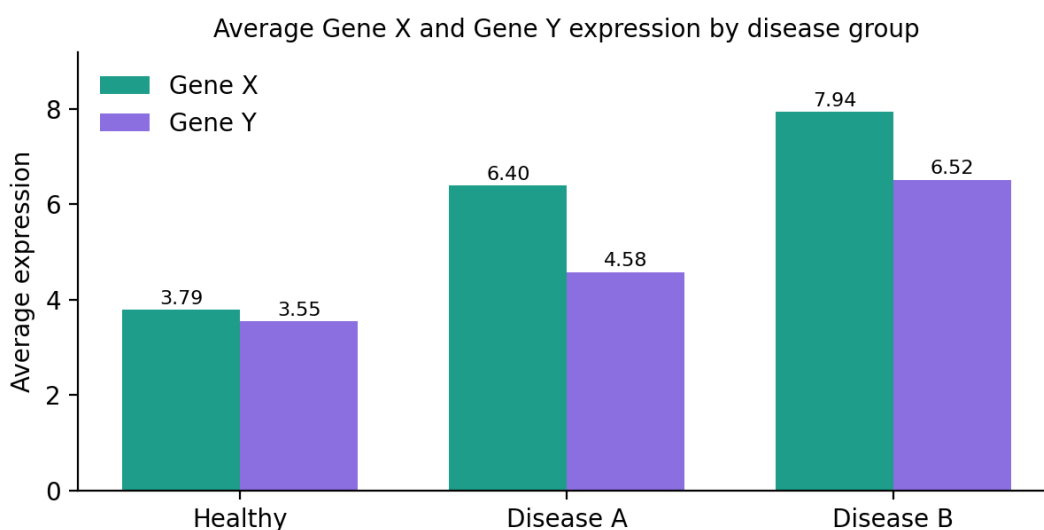


Figure 1: Average gene expression by disease group

4.2 Smoking status and disease

A strong pattern was observed between smoking status and disease status in this dataset: non-smokers corresponded to the Healthy group, ex-smokers to Disease A, and current smokers to Disease B.

4.3 Treatment response

Average expression differed across treatment-response categories. Full Response had the lowest average expression, Partial Response was intermediate, and No Response had the highest. Treatment-response categories also aligned strongly with disease categories.

Because the data is observational and the disease groups are small (18 and 20 patients), these results describe associations within this dataset and do not show cause and effect.

4.4 Dashboard



Figure 2: Power BI dashboard

5. Conclusion

The project shows a complete analytics workflow, from raw data to a shareable dashboard. Gene X and Gene Y expression are higher in the disease groups, with Disease B highest, and strong patterns link disease status with smoking status and treatment response. Possible next steps are statistical significance testing and a larger dataset with more disease cases.